Carelink – Non-Profit Quiz/Tutor Bot

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Academic Year: 2025-26

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1. Problem Statement & Objectives

1.1 Problem Statement

Non-profit organizations receive a large number of messages related to donations, emergency help, and general support. Managing these messages manually leads to delays, misclassification, and inefficient prioritization. Critical messages may get ignored due to lack of urgency detection.

The absence of an intelligent system results in poor response time, increased manual workload, and lack of structured data handling. Therefore, an AI-based solution is required to automate message classification and improve overall efficiency of communication systems.

1.2 Project Objectives

The CARELINK system is designed to achieve multiple objectives that improve communication handling in non-profit organizations.

The main objectives of the project are to automatically classify messages based on urgency levels such as High, Medium, and Low. The system uses AI techniques to analyze incoming messages and generate appropriate responses. It also provides a centralized dashboard where administrators can view and manage messages effectively.

Additionally, the system includes an interactive chatbot to assist users and aims to reduce manual effort while improving response efficiency and accuracy.

1.3 Scope of the Project

The scope of this project includes the development of an AI-powered system that processes and classifies messages for better decision-making. The system focuses on building a user-friendly frontend interface, a backend API for handling requests, and an AI module for classification.

The project is limited to core functionalities such as message processing, dashboard visualization, and chatbot interaction. Advanced features like full-scale deployment, payment integration, and large-scale cloud optimization are not included in the current scope

2. Proposed Solution

The CARELINK system is designed as an intelligent platform that automates the process of handling and analyzing messages for non-profit organizations. It integrates artificial intelligence with a structured workflow to ensure efficient communication and faster decision-making.

The system processes user messages through a frontend interface, sends them to a backend API, and applies AI-based analysis to classify urgency levels. The results are then displayed on a dashboard where administrators can monitor and take appropriate actions. This approach minimizes manual effort and ensures that critical messages are handled with priority.

2.1 Key Features

The CARELINK system provides several important features that enhance the efficiency of communication management.

The system includes automated urgency classification, where messages are categorized into High, Medium, or Low priority using AI-based logic. This helps in identifying urgent cases quickly. Another key feature is AI-generated responses, which assist administrators in replying to messages without manually typing every response.

A centralized dashboard is provided to monitor all incoming messages and view their status. The system also includes an interactive chatbot (Quiz Tutor Bot) that engages with users and provides instant support. Additionally, the platform offers a clean and user-friendly interface along with basic analytics to track message trends.

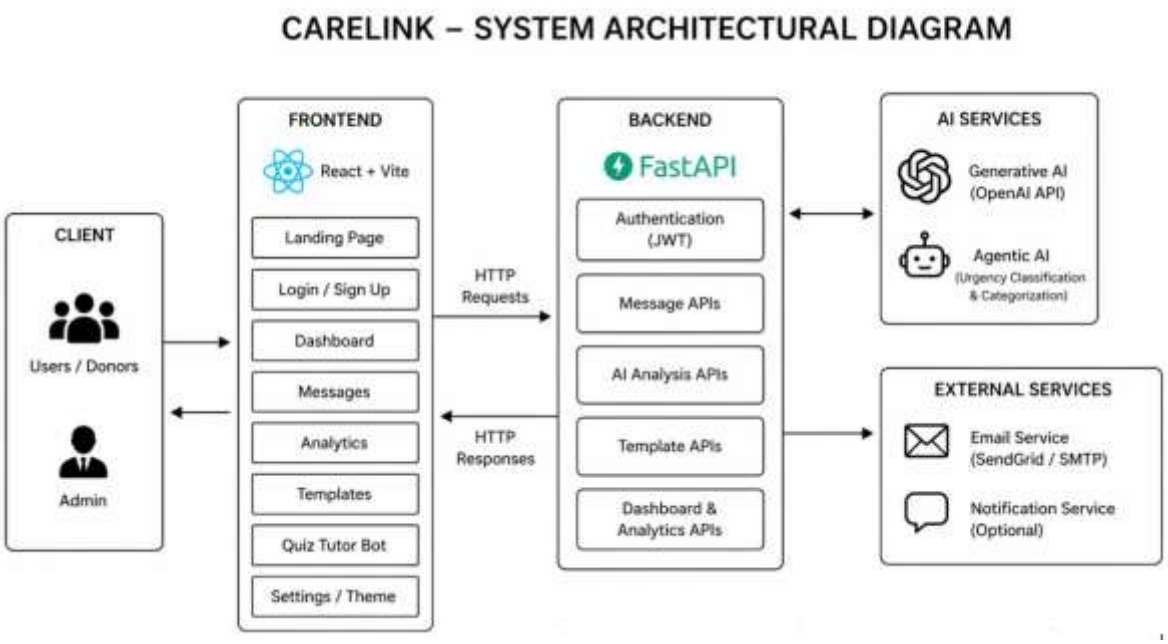
These features collectively improve response time, reduce workload, and ensure better management of communication processes.

2.2 Overall Architecture / Workflow

The CARELINK system follows a structured architecture where each component performs a specific role in processing user messages.

The workflow begins when a user submits a message through the frontend interface. This message is sent to the backend API, which processes the request and forwards it to the AI module. The AI module analyzes the message content and classifies it based on urgency. The processed result is then stored and displayed on the dashboard for the administrator.

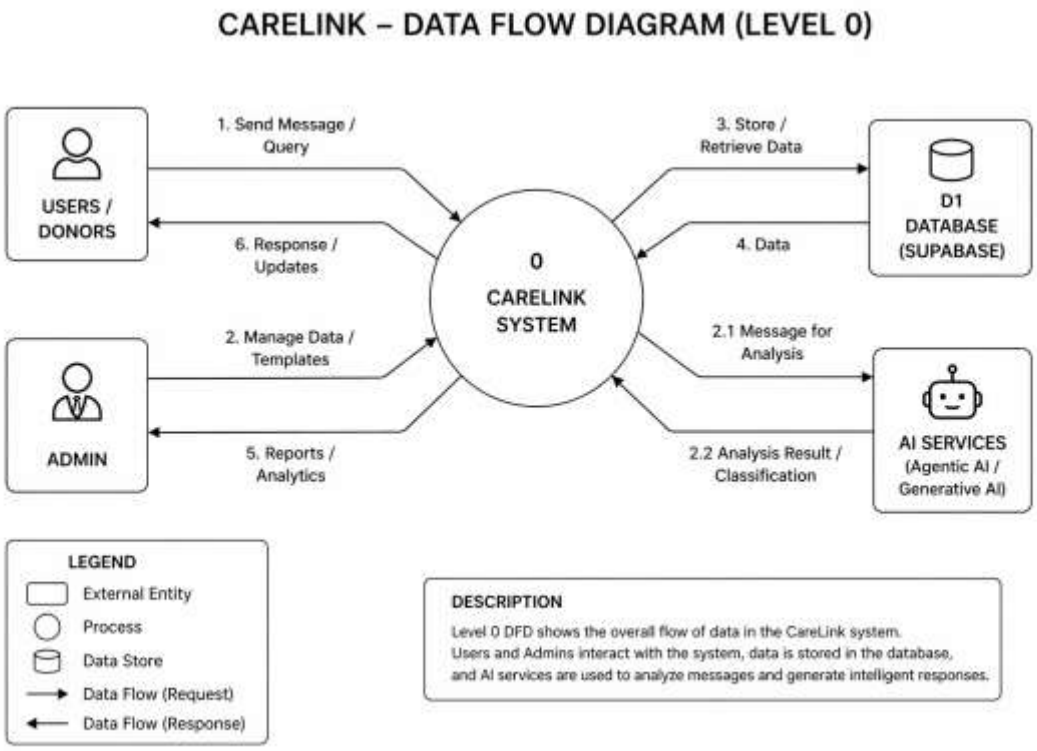
This architecture ensures smooth data flow and efficient processing of information across different components of the system.



2.2.1 Data Flow Diagram (DFD)

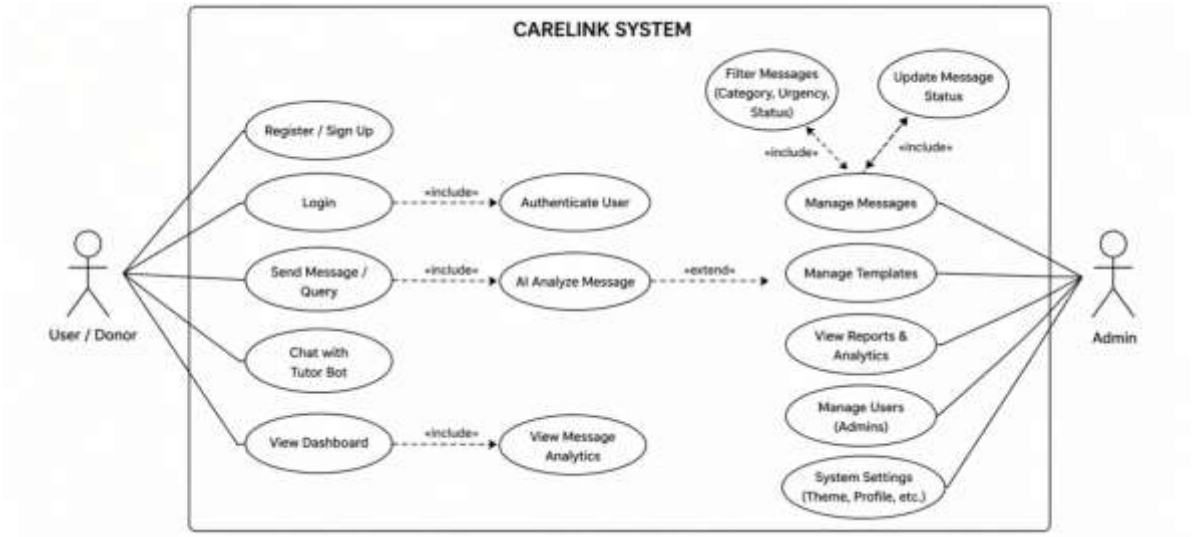
DFD Level 0

The Level 0 Data Flow Diagram provides a high-level overview of the system. It shows the interaction between the user and the CARELINK system, where the user sends messages and receives processed outputs.



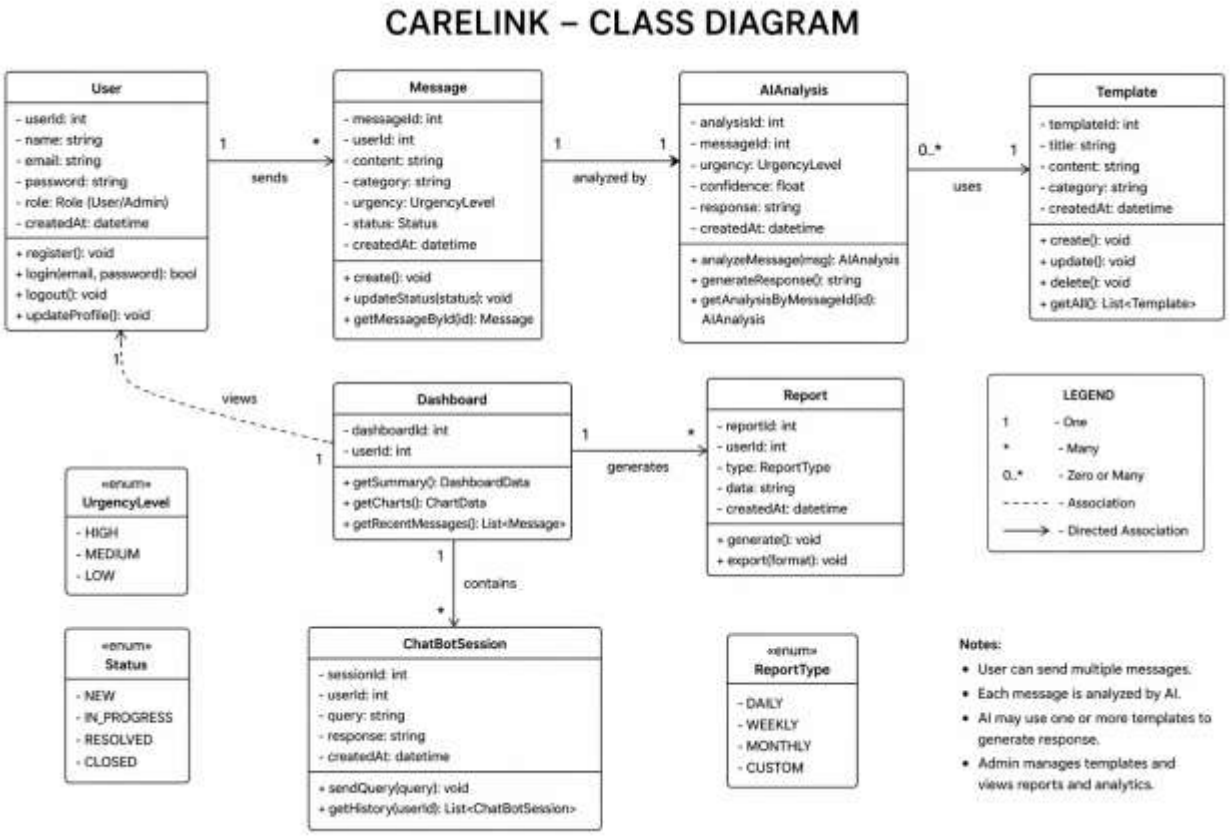
2.2.2 Use Case Diagram

The Use Case Diagram represents the interaction between users and the system. It highlights the functionalities such as sending messages, viewing the dashboard, analyzing messages, and managing system data.



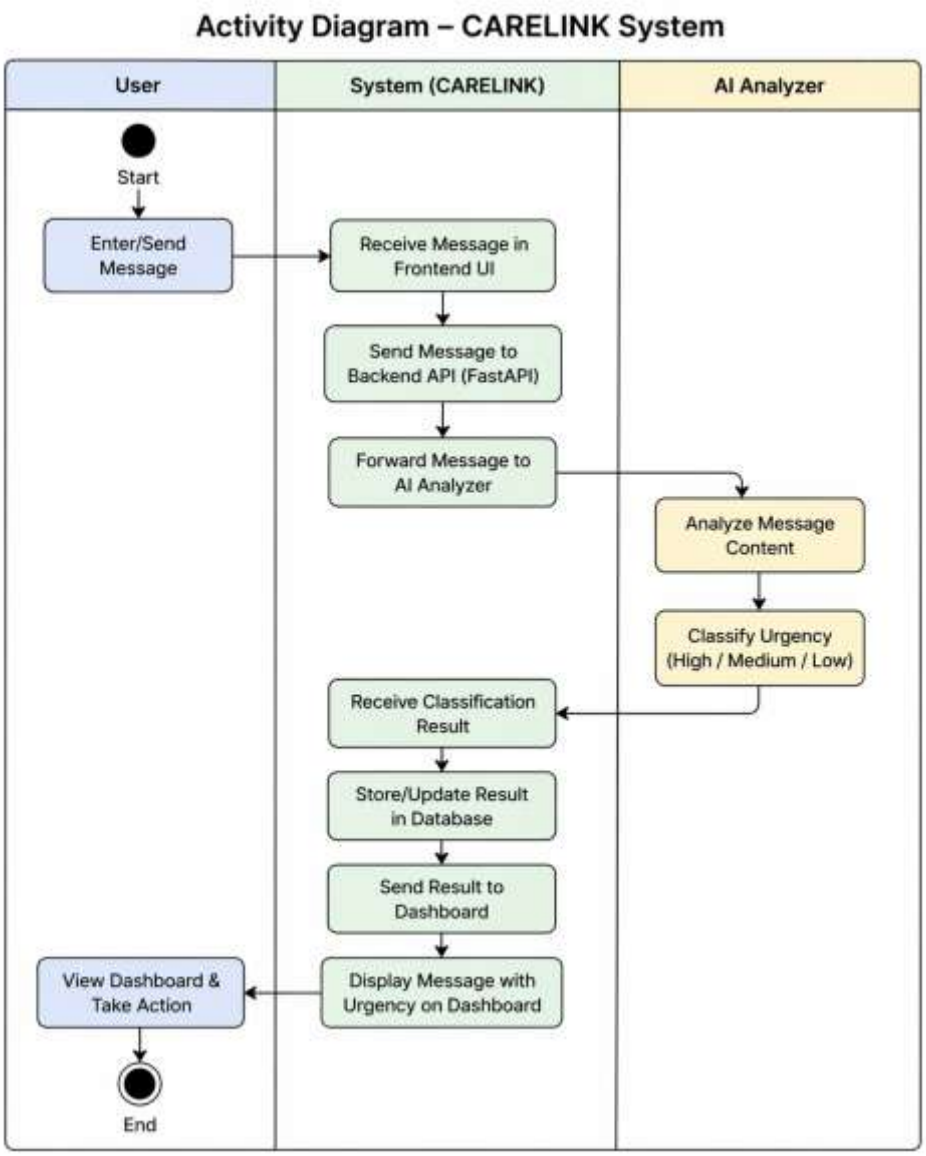
2.2.3 Class Diagram

The Class Diagram represents the structure of the system by showing different classes such as User, Message, AI Analyzer, and Dashboard along with their relationships.



2.2.4 Activity Diagram

The Activity Diagram illustrates the step-by-step workflow of the system, starting from user input to message processing, classification, and final output display.



2.3 Tools & Technologies Used

The CARELINK system uses a combination of modern tools and technologies to ensure efficient performance and scalability.

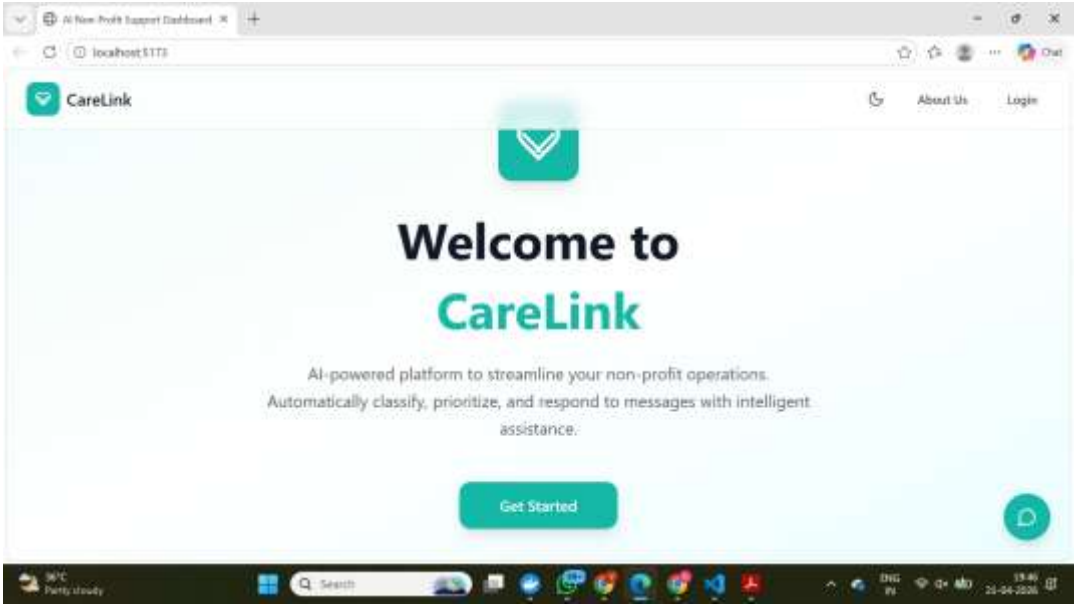
Category	Tools Used
Frontend	React, Vite
Backend	FastAPI
AI	OpenAI / Rule-based
Database	Supabase
Styling	Tailwind CSS
Version Control	Git, GitHub

3. Results & Output

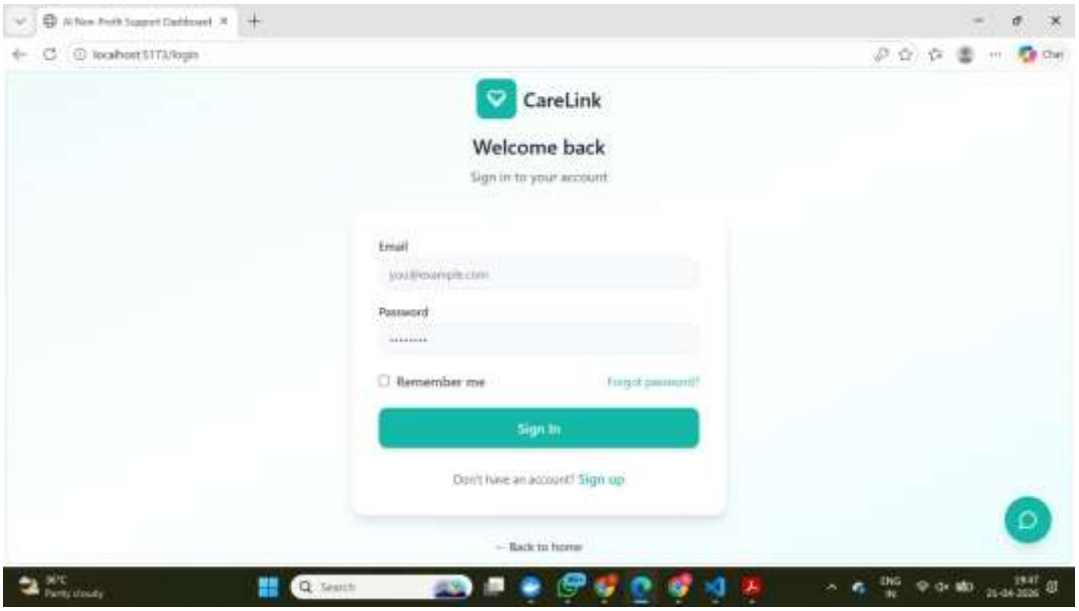
This section presents the outputs generated by the CARELINK system along with screenshots and system results. It demonstrates how the system performs message classification and displays results through a user-friendly dashboard.

3.1 Outputs (Screenshots)

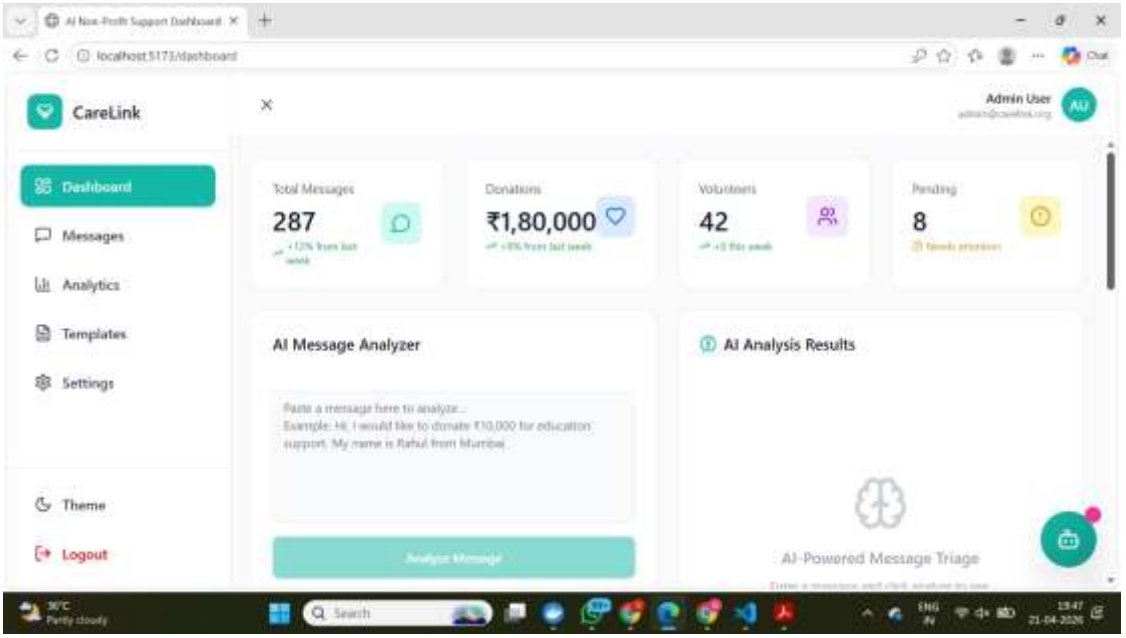
1. Home Page



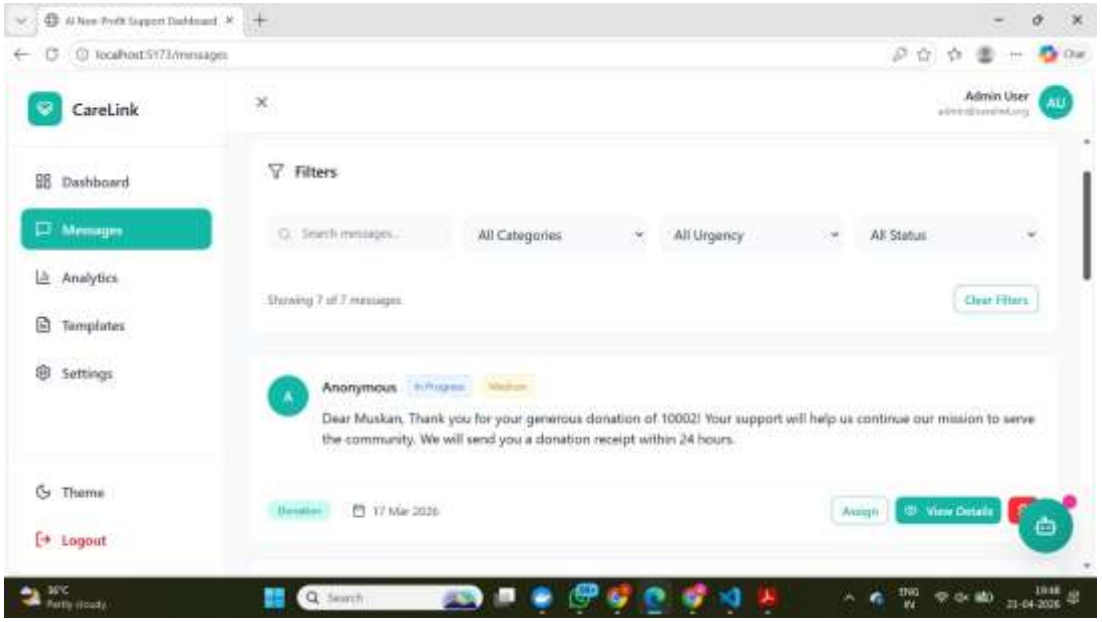
2. login page



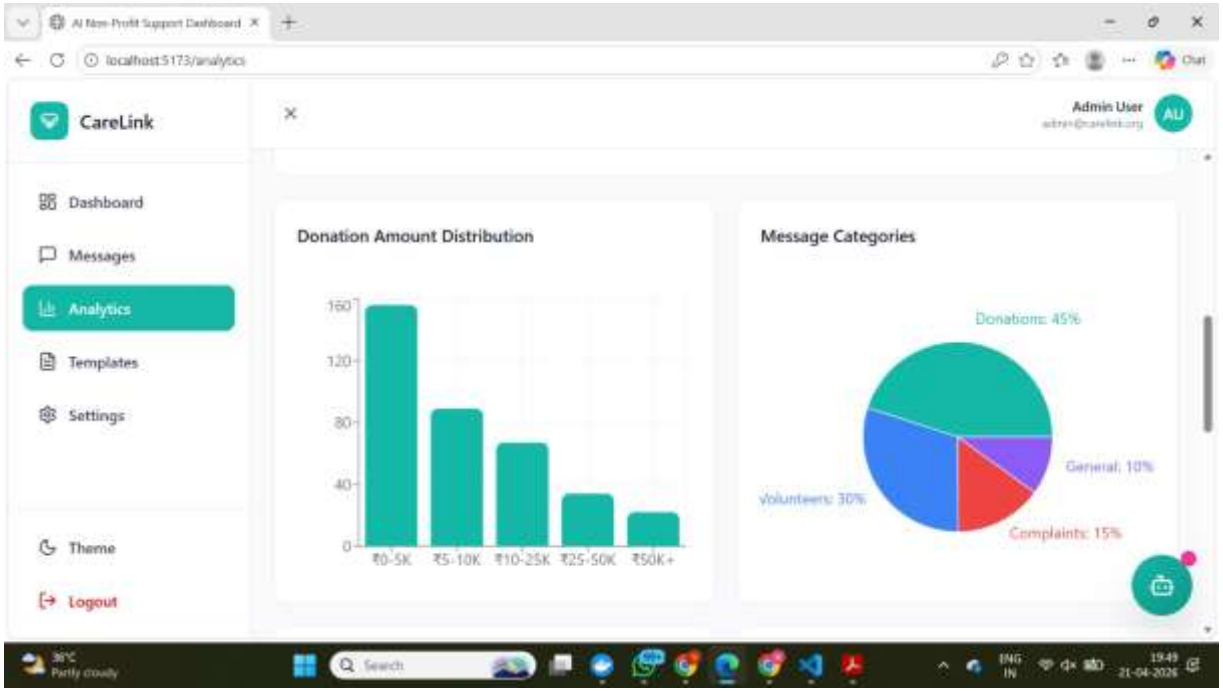
3. Dashboard



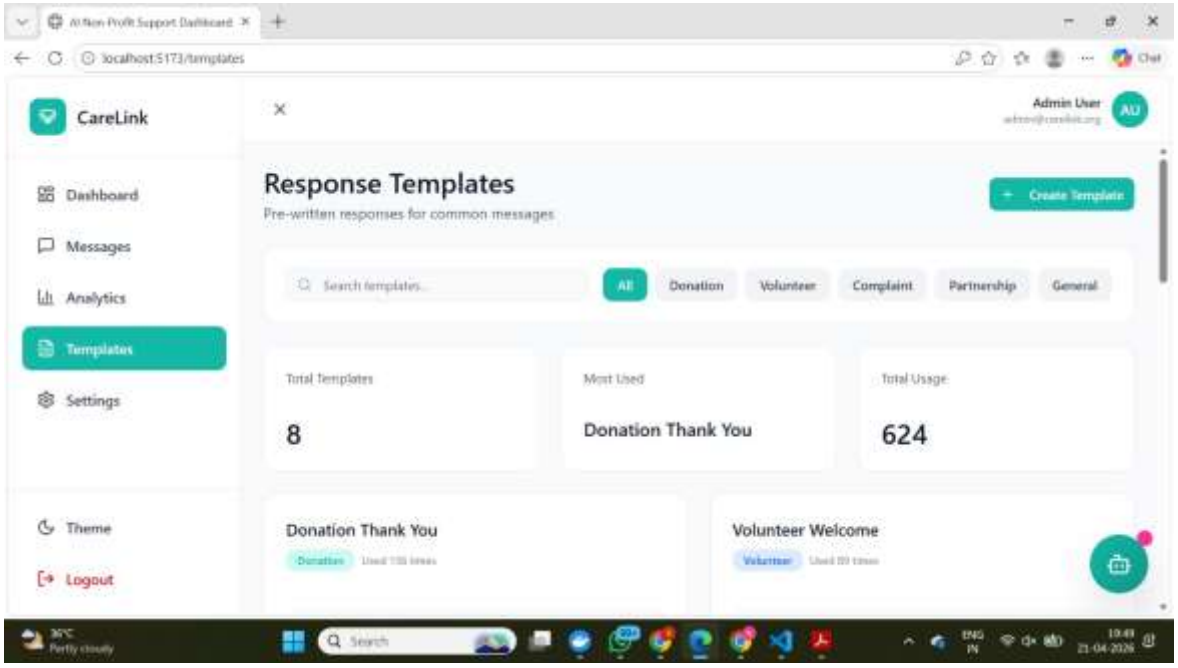
4. Message Management



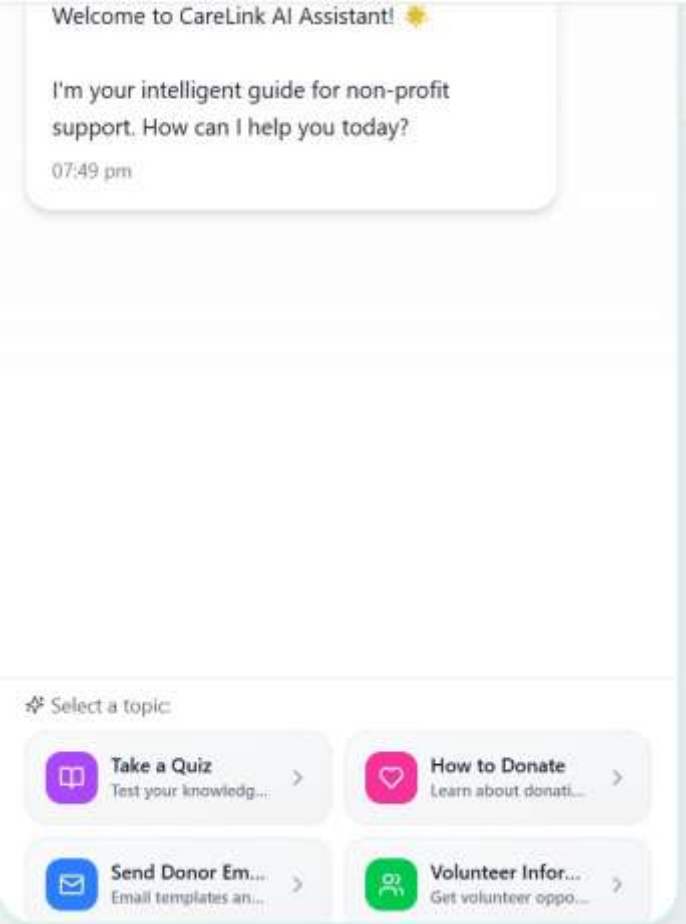
5. Analytics



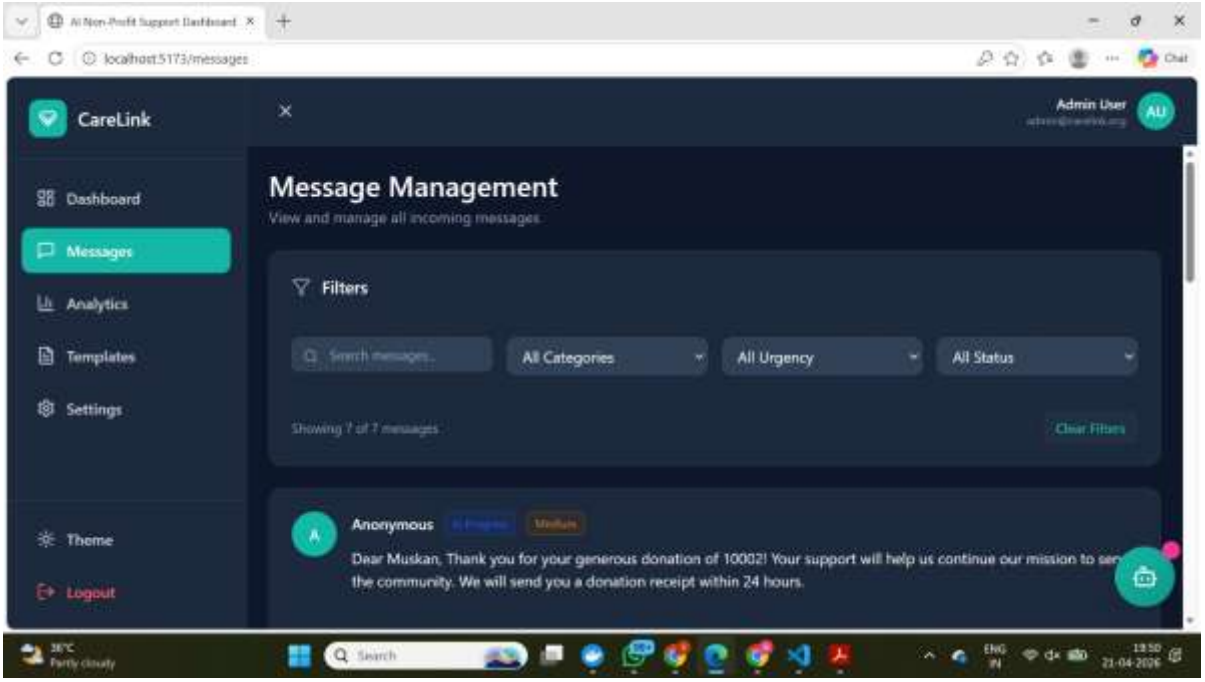
6. Templates



7. Quiz Tutor Bot



8. Theme Feature



3.2 Reports (Dashboards / Models)

The CARELINK system generates reports through its dashboard which helps in analyzing system performance. These reports include information about message categories, urgency levels, and response patterns.

The dashboard provides visual representation using charts and graphs, making it easier for administrators to understand trends and take necessary actions. The AI model also contributes by providing classification results that are reflected in these reports.

3.3 Key Outcomes

The implementation of the CARELINK system results in several important outcomes.

The system successfully automates the process of message classification, reducing the need for manual effort. It improves response time by prioritizing urgent messages and ensures that critical issues are handled promptly.

Additionally, the dashboard provides clear insights into message trends, helping in better decision-making. Overall, the system enhances efficiency, accuracy, and communication management in non-profit organizations.

4. Conclusion

The CARELINK system successfully demonstrates the use of artificial intelligence in automating message handling for non-profit organizations. The system is capable of receiving user messages, processing them through a backend API, and analyzing them using an AI module to determine urgency levels.

By integrating a structured workflow and a centralized dashboard, the system improves response time and reduces manual effort. It ensures that important messages are identified and handled efficiently. The user-friendly interface and chatbot functionality further enhance the overall usability of the system.

In conclusion, CARELINK provides an effective and practical solution for managing communication in a more organized, accurate, and efficient manner.

5. Future Scope & Enhancements

The CARELINK system can be enhanced by integrating advanced AI models for more accurate message classification. Real-time notifications such as email or SMS alerts can be added for urgent messages. A mobile application can improve accessibility, and multi-language support can make the system usable for a wider audience. Additionally, cloud deployment and scalability improvements can help handle larger data efficiently.